Compactness through finite intersections

Section 3: Products

All levels · read along the `refines` order

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3 Products

Finite products of compact spaces are compact. The tube lemma (Lemma 2.8) is stated in Section 2.2*; it is the whole content of the argument.

Theorem 3.1 (Products of two compact spaces)**.** *If X and Y are compact, so is $X \times Y$ with the product topology.*

Proof. Let $\mathcal{W}$ be an open cover of $X \times Y$. For each $x \in X$ the slice $\{x\} \times Y \cong Y$ is compact, so finitely many members of $\mathcal{W}$ cover it; call their union W_x. By Lemma 2.8 there is an open $U_x \ni x$ with $U_x \times Y \subseteq W_x$.

1. The U_x cover X, so by Definition 2.3* finitely many do: $U_{x_1}, \dots, U_{x_m}$.
2. Then $X \times Y = \bigcup_{k=1}^{m} U_{x_k} \times Y \subseteq \bigcup_k W_{x_k}$, a union of finitely many members of $\mathcal{W}$.

Compare the closed-set form, Theorem 2.4*, which gives the same result by a less direct route. ∎

Remark 3.2 (Arbitrary products)**.** For an arbitrary family $(X_\alpha)_{\alpha \in A}$ the statement is Tychonoff's theorem, which needs the axiom of choice; the argument above only gives finite products, by induction on the number of factors: $\prod_{k=1}^{n+1} X_k \cong (\prod_{k=1}^{n} X_k) \times X_{n+1}$.

Written out here; in the full blueprint it is read under 2.2 Compact spaces, outside this export.

Lemma 2.8 (Tube lemma)**.** *Let Y be compact, $x \in X$, and $W \subseteq X \times Y$ open with $\{x\} \times Y \subseteq W$. Then there is an open $U \ni x$ with $U \times Y \subseteq W$.*

Proof. Cover $\{x\} \times Y$ by basic open sets $U_y \times V_y \subseteq W$. The V_y cover Y, so finitely many do, say $V_{y_1}, \dots, V_{y_n}$, and $U = U_{y_1} \cap \dots \cap U_{y_n}$ is the open set wanted. ∎

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References marked * are to parts of the blueprint this document leaves out:

Section 2.2 Compact spaces, in 2 Covers and compact spaces (`compact`)
Definition 2.3 Compact space, in 2 Covers and compact spaces (`def-compact`)
Theorem 2.4 Compactness by closed sets, in 2 Covers and compact spaces (`thm-fip`)